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**Sujet :** systemd, Réseau Bond/VLAN & Automatisation

**Hôtes :** srv-core (Rocky Linux 9.8) / srv-edge

**Synthèse Exécutive**

- **Optimisation système :** Audit du temps de démarrage avec systemd-analyze.
- **Sécurité applicative :** Déploiement et durcissement du service helios-collect.service (score d'exposition réduit de **5.3 MEDIUM** à **1.4 OK**).
- **Haute Disponibilité Réseau :** Implémentation d'une agrégation de liens bond0 (Active-Backup), segmentation VLAN 42 et routage par règles (PBR).
- **Automatisation :** Création d'un script d'archivage sécurisé piloté par systemd.timer.

**1. Services, Processus et Séquence de Démarrage**

**1.1 Séquence de démarrage (`systemd-analyze`)**

Bilan du démarrage initial et tentative d'export du graphique de boot :

```
noah@192.168.56.110's password:
Last login: Mon Nov  2 14:10:02 2026
[noah@localhost ~]$ systemd-analyze blame | head -15
systemd-analyze critical-chain
systemd-analyze
systemd-analyze plot > /root/boot-avant.svg
14.675s dnf-makecache.service
 5.985s NetworkManager-wait-online.service
 5.188s kdump.service
 5.010s dev-ttyS3.device
 5.010s sys-devices-platform-serial8250-tty-ttyS3.device
 5.009s sys-devices-platform-serial8250-tty-ttyS2.device
 5.009s dev-ttyS2.device
 5.007s sys-devices-platform-serial8250-tty-ttyS1.device
 5.007s dev-ttyS1.device
 5.003s dev-ttyS0.device
 5.003s sys-devices-platform-serial8250-tty-ttyS0.device
 4.998s sys-module-configfs.device
 4.656s dev-disk-by\x2dpartuuid-b46f2970\x2d02.device
 4.656s sys-devices-pci0000:00-0000:00:0d.0-ata3-host2-target2:0:0-2:0:0:0-b
lock-sda-sda2.device
 4.656s dev-sda2.device
The time when unit became active or started is printed after the "@" charac>
The time the unit took to start is printed after the "+" character.

multi-user.target @9.979s
└─getty.target @4.117s
 ESCOC
arted is printed after the "@" character.
```

**1.2 Création du script de collecte sous root**

Passage en privilèges superutilisateur (`sudo -i`) et écriture du script d'arrière-plan `/usr/local/sbin/helios-collect.sh` :

```
Startup finished in 1.838s (kernel) + 4.382s (initrd) + 15.097s (userspace)
= 21.318s
multi-user.target reached after 9.979s in userspace.
-bash: /root/boot-avant.svg: Permission non accordée
[noah@localhost ~]$ sudo -i
[sudo] Mot de passe de noah :
[root@localhost ~]# install -d -m 0755 /usr/local/sbin /var/lib/helios /var/
log/helios
[root@localhost ~]# cat > /usr/local/sbin/helios-collect.sh <<'EOF'
#!/usr/bin/bash
# Collecteur de telemetrie Helios boucle longue duree
set -euo pipefail

INTERVAL="${HELIOS_INTERVAL:-15}"
OUT=/var/lib/helios/telemetry.csv

while :; do
    printf '%s, %s, %s\n' "$(date --iso-8601=seconds)" \
                          "$(awk '{print $1}' /proc/loadavg)" \
                          "$(awk '/MemAvailable/ {print $2}' /proc/meminfo)"
 >> "$OUT"
    sleep "$INTERVAL"
done
EOF

chmod 0755 /usr/local/sbin/helios-collect.sh
```

### 1.3 Définition initiale et score d'exposition intermédiaire

Contrôle de l'état des interfaces réseau et statut initial du service :

```
WantedBy=multi-user.target
-bash: [Install] : commande introuvable
[root@localhost ~]# CapabilityBoundingSet=
[root@localhost ~]# WantedBy=multi-user.target
[root@localhost ~]# # srv-core, en root (sudo -i)
ip -br addr ; ip -br link        # note le nom de tes 2 interfaces helios-la
n
cat /etc/os-release | head -2    # Rocky Linux 9
systemctl is-active NetworkManager firewalld
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP             10.0.2.15/24 fd17:625c:f037:2:a00:27ff:feb8:
8186/64 fe80::a00:27ff:feb8:8186/64
enp0s8           UP             192.168.56.110/24 fe80::4a48:4254:fbc5:9635/
64
enp0s9           DOWN
enp0s10          DOWN
lo               UNKNOWN        00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
enp0s3           UP             08:00:27:b8:81:86 <BROADCAST,MULTICAST,UP,LO
WER_UP>
enp0s8           UP             08:00:27:9e:a9:ce <BROADCAST,MULTICAST,UP,LO
WER_UP>
enp0s9           DOWN           08:00:27:18:46:c4 <NO-CARRIER,BROADCAST,MULT
ICAST,UP>
enp0s10          DOWN           08:00:27:16:5e:c4 <NO-CARRIER,BROADCAST,MULT
ICAST,UP>
NAME="Rocky Linux"
VERSION="9.8 (Blue Onyx)"
active
inactive
[root@localhost ~]#
```

### 1.4 Durcissement de sécurité du service systemd

Application des directives d'isolation avancées via `sed` et vérification du score :

```
→ Overall exposure level for helios-collect.service: 5.3 MEDIUM 😬
lines 44-83/83 (END)
```

```
[root@localhost ~]# systemctl cat helios-collect.service --no-pager | tail -
8
ReadWritePaths=/var/lib/helios /var/log/helios
ProtectKernelTunables=yes
ProtectControlGroups=yes
RestrictSUIDSGID=yes
CapabilityBoundingSet=

[Install]
WantedBy=multi-user.target
[root@localhost ~]#
```

**1.5 Test de résilience (Kill -9 et Auto-Restart)**

Simulation d'un arrêt brutal du PID principal et contrôle de la relance dans `journalctl` :

```
WantedBy=multi-user.target
[root@localhost ~]# ProtectKernelModules=yes
ProtectKernelLogs=yes
ProtectClock=yes
ProtectHostname=yes
PrivateDevices=yes
PrivateNetwork=yes
RestrictNamespaces=yes
RestrictRealtime=yes
LockPersonality=yes
MemoryDenyWriteExecute=yes
SystemCallArchitectures=native
SystemCallFilter=@system-service
RestrictAddressFamilies=AF_UNIX
[root@localhost ~]# systemctl is-active helios-collect.service
tail -3 /var/lib/helios/telemetry.csv
active
2026-11-02T14:35:33+01:00, 0.00, 3198084
2026-11-02T14:35:48+01:00, 0.00, 3198448
2026-11-02T14:36:03+01:00, 0.00, 3198332
```

**1.6 Cible métier `helios.target` & Ordonnancement CPU/IO**

Création de la target, liaison des dépendances et modification des priorités dynamiques:

```
→ Overall exposure level for helios-collect.service: 5.3 MEDIUM 😐
[root@localhost ~]# sed -i '/^\[Install\]/i \
ProtectKernelModules=yes\
ProtectKernelLogs=yes\
ProtectClock=yes\
ProtectHostname=yes\
PrivateDevices=yes\
RestrictNamespaces=yes\
RestrictRealtime=yes\
LockPersonality=yes\
MemoryDenyWriteExecute=yes\
SystemCallArchitectures=native\
SystemCallFilter=@system-service\
RestrictAddressFamilies=AF_UNIX\
' /etc/systemd/system/helios-collect.service

systemctl daemon-reload && systemctl restart helios-collect.service
systemctl is-active helios-collect.service
systemd-analyze security helios-collect.service --no-pager | tail -3
active
✗ UMask=                                                   Files created
by service are world-readable by default
               0.1

→ Overall exposure level for helios-collect.service: 1.4 OK 🙂
[root@localhost ~]#
```

### 1.7 Inspection des tranches Control Groups (systemd-cgls / systemd-cgtop)

Relevé des cgroups système et des processus actifs :

```
[root@localhost ~]# kill -9 "$(systemctl show -p MainPID --value helios-coll
ect.service)"
sleep 8 ; systemctl status helios-collect.service --no-pager | head -5
● helios-collect.service - Collecteur de telemetrie Helios
     Loaded: loaded (/etc/systemd/system/helios-collect.service; enabled; pr
eset: disabled)
     Active: active (running) since Mon 2026-11-02 14:40:19 CET; 2s ago
       Docs: https://wiki.helios.lan/collect
   Main PID: 45736 (helios-collect.)
[root@localhost ~]#
```

## 2. Réseau Avancé : Bond, VLAN & Routage PBR

### 2.1 Agrégation de liens (`bond0` Mode Active-Backup)

Vérification du statut du driver de bonding après configuration sur `srv-core` :

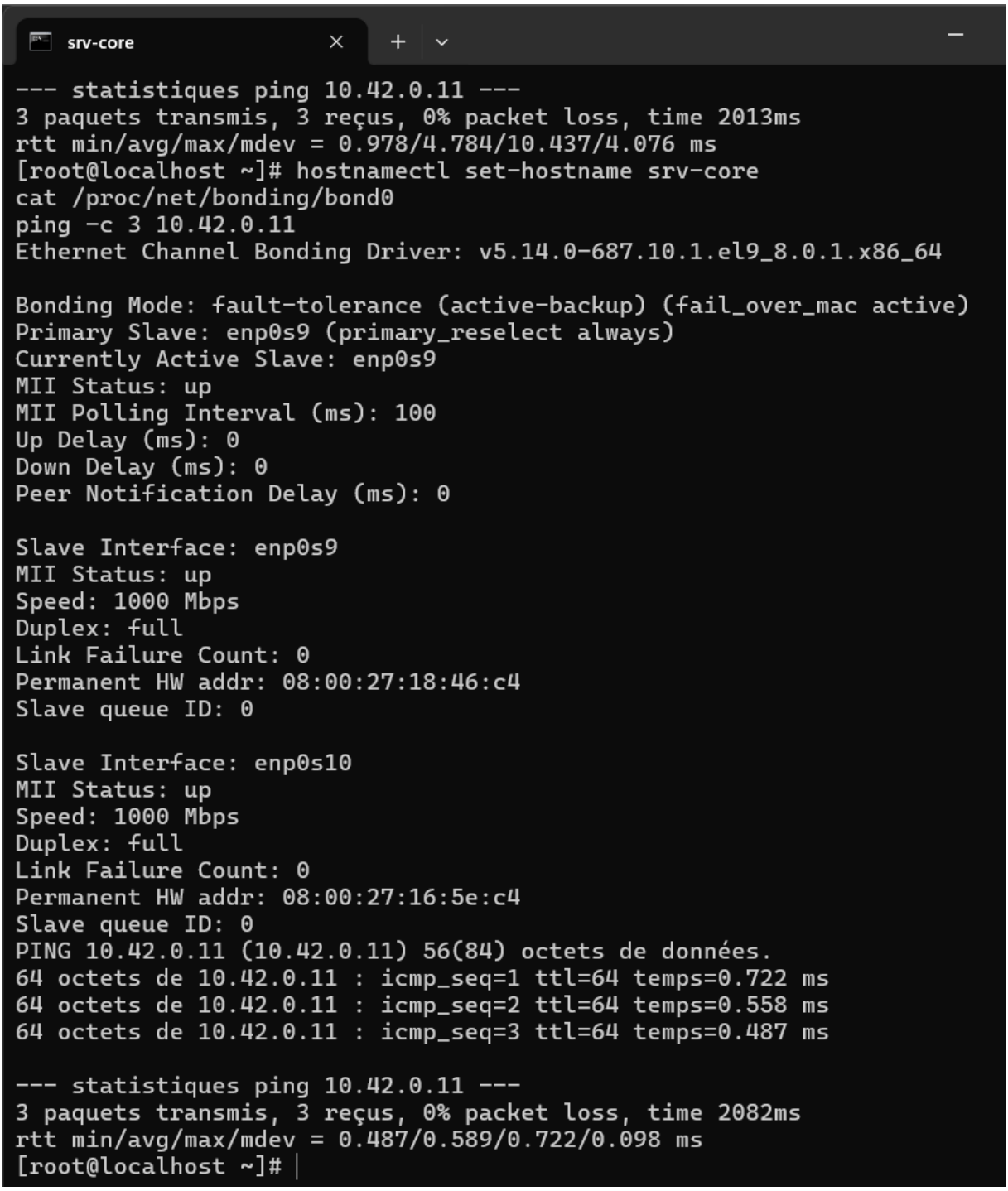
```
[root@localhost ~]# systemctl show -p MainPID --value helios-collect.service
   # avant
# kill -9 ...
systemctl show -p MainPID --value helios-collect.service   # après (différen
t)
journalctl -u helios-collect.service -n 10 --no-pager      # on y voit le SI
GKILL puis le redémarrage
45736
45736
nov. 02 14:39:30 localhost.localdomain systemd[1]: Started Collecteur de tel
emetrie Helios.
nov. 02 14:39:42 localhost.localdomain systemd[1]: Stopping Collecteur de te
lemetrie Helios...
nov. 02 14:39:42 localhost.localdomain systemd[1]: helios-collect.service: D
eactivated successfully.
nov. 02 14:39:42 localhost.localdomain systemd[1]: Stopped Collecteur de tel
emetrie Helios.
nov. 02 14:39:42 localhost.localdomain systemd[1]: Started Collecteur de tel
emetrie Helios.
nov. 02 14:40:14 localhost.localdomain systemd[1]: helios-collect.service: M
ain process exited, code=killed, status=9/KILL
nov. 02 14:40:14 localhost.localdomain systemd[1]: helios-collect.service: F
ailed with result 'signal'.
nov. 02 14:40:19 localhost.localdomain systemd[1]: helios-collect.service: S
cheduled restart job, restart counter is at 1.
nov. 02 14:40:19 localhost.localdomain systemd[1]: Stopped Collecteur de tel
emetrie Helios.
nov. 02 14:40:19 localhost.localdomain systemd[1]: Started Collecteur de tel
emetrie Helios.
[root@localhost ~]#
```

```
systemctl daemon-reload
systemctl add-wants helios.target helios-collect.service
systemctl enable helios.target
systemctl list-dependencies helios.target
systemctl stop helios.target && systemctl start helios.target
Created symlink /etc/systemd/system/helios.target.wants/helios-collect.servi
ce → /etc/systemd/system/helios-collect.service.
Created symlink /etc/systemd/system/multi-user.target.wants/helios.target →
/etc/systemd/system/helios.target.
helios.target
● ├─helios-collect.service
● └─multi-user.target
●   ├─auditd.service
●   ├─chronyd.service
●   ├─crond.service
●   ├─helios-collect.service
●   ├─irqbalance.service
●   ├─kdump.service
●   ├─NetworkManager.service
●   ├─rsyslog.service
●   ├─sshd.service
```

### 2.2 Test de Bascule en charge (Failover avec coupure de lien)

Test de bascule sous flux de ping continu vers `10.42.0.11` :

```
[root@localhost ~]# nice -n 19 sha256sum /dev/zero & NPID=$!
ps -o pid,ni,pri,cls,rtprio,comm -p "$NPID"
renice -n 5 -p "$NPID" ; ionice -c 2 -n 7 -p "$NPID" ; ionice -p "$NPID" ; c
hrt -p "$NPID"
kill "$NPID"
[1] 46017
    PID  NI PRI CLS RTPRIO COMMAND
  46017  19   0  TS      - sha256sum
46017 (process ID) priorité précédente 19, nouvelle priorité 5
best-effort : priorité 7
stratégie d'ordonnancement actuelle pour le PID 46017 : SCHED_OTHER
priorité de planification actuelle pour le PID 46017 : 0
[root@localhost ~]# systemctl show -p MemoryMax ...
MemoryMax=infinity
[1]+  Complété                nice -n 19 sha256sum /dev/zero
[root@localhost ~]# list-dependencies
-bash: list-dependencies : commande introuvable
[root@localhost ~]#
user.slice                              11    1,6     1.6G        -        -
user.slice/user-1000.slice              11    1,6     1.6G        -        -
user.slice/….slice/session-5.scope       7    1,6     8.1M        -        -
system.slice                            20    0,0   205.5M        -        -
system.slice/rsyslog.service             3    0,0     2.1M        -        -
dev-hugepages.mount                      -      -    52.0K        -        -
dev-mqueue.mount                         -      -     4.0K        -        -
init.scope                               1      -    33.9M        -        -
sys-fs-fuse-connections.mount            -      -     4.0K        -        -
sys-kernel-config.mount                  -      -     4.0K        -        -
sys-kernel-debug.mount                   -      -     4.0K        -        -
sys-kernel-tracing.mount                 -      -     4.0K        -        -
system.slic…NetworkManager.service       3      -    10.5M        -        -
system.slice/auditd.service              2      -     4.0M        -        -
system.slice/boot.mount                  -      -     4.0K        -        -
system.slice/chronyd.service             1      -     5.1M        -        -
system.slice/crond.service               1      -  1000.0K        -        -
system.slice/dbus-broker.service         2      -     2.3M        -        -
system.slic…apper-rlm\x2dswap.swap       -      -   164.0K        -        -
system.slic…helios-collect.service       2      -   784.0K        -        -
system.slice/irqbalance.service          2      -     1.0M        -        -
system.slice/sshd.service                1      -     4.0M        -        -
system.slice/sysroot.mount               -      -     4.0K        -        -
system.slice/system-getty.slice          -      -     2.8M        -        -
system.slic…ice/getty@tty1.service       -      -     1.4M        -        -
system.slice/system-modprobe.slice       -      -   656.0K        -        -
system.slic…ernate\x2dresume.slice       -      -     4.0K        -        -
system.slic…stemd-journald.service       1      -     1.5M        -        -
system.slic…systemd-logind.service       1      -     4.7M        -        -
system.slice/systemd-udevd.service       1      -     3.2M        -        -
user.slice/….slice/session-3.scope       2      -     1.6G        -        -
user.slice/…lice/user@1000.service       2      -     7.8M        -        -
enabled
helios.target
● ├─helios-collect.service
● └─multi-user.target
●   ├─auditd.service
[root@localhost ~]#
```


srv-core
--- statistiques ping 10.42.0.11 ---
3 paquets transmis, 3 reçus, 0% packet loss, time 2013ms
rtt min/avg/max/mdev = 0.978/4.784/10.437/4.076 ms
[root@localhost ~]# hostnamectl set-hostname srv-core
cat /proc/net/bonding/bond0
ping -c 3 10.42.0.11
Ethernet Channel Bonding Driver: v5.14.0-687.10.1.el9_8.0.1.x86_64
Bonding Mode: fault-tolerance (active-backup) (fail_over_mac active)
Primary Slave: enp0s9 (primary_reselect always)
Currently Active Slave: enp0s9
MII Status: up
MII Polling Interval (ms): 100
Up Delay (ms): 0
Down Delay (ms): 0
Peer Notification Delay (ms): 0
Slave Interface: enp0s9
MII Status: up
Speed: 1000 Mbps
Duplex: full
Link Failure Count: 0
Permanent HW addr: 08:00:27:18:46:c4
Slave queue ID: 0
Slave Interface: enp0s10
MII Status: up
Speed: 1000 Mbps
Duplex: full
Link Failure Count: 0
Permanent HW addr: 08:00:27:16:5e:c4
Slave queue ID: 0
PING 10.42.0.11 (10.42.0.11) 56(84) octets de données.
64 octets de 10.42.0.11 : icmp_seq=1 ttl=64 temps=0.722 ms
64 octets de 10.42.0.11 : icmp_seq=2 ttl=64 temps=0.558 ms
64 octets de 10.42.0.11 : icmp_seq=3 ttl=64 temps=0.487 ms
--- statistiques ping 10.42.0.11 ---
3 paquets transmis, 3 reçus, 0% packet loss, time 2082ms
rtt min/avg/max/mdev = 0.487/0.589/0.722/0.098 ms
[root@localhost ~]#

### 2.3 Configuration du VLAN 802.1Q (VLAN 42)

Création des sous-interfaces étiquetées sur `srv-core` (`bond0.42`) et `srv-edge` (`vlan42`) :

```
he new active one
[root@localhost ~]# ping -O 10.42.0.11 | tee /root/ping-bascule.txt
PING 10.42.0.11 (10.42.0.11) 56(84) octets de données.
64 octets de 10.42.0.11 : icmp_seq=1 ttl=64 temps=3.06 ms
64 octets de 10.42.0.11 : icmp_seq=2 ttl=64 temps=0.780 ms
64 octets de 10.42.0.11 : icmp_seq=3 ttl=64 temps=0.814 ms
64 octets de 10.42.0.11 : icmp_seq=4 ttl=64 temps=1.66 ms
64 octets de 10.42.0.11 : icmp_seq=5 ttl=64 temps=0.553 ms
64 octets de 10.42.0.11 : icmp_seq=6 ttl=64 temps=1.30 ms
no answer yet for icmp_seq=7
no answer yet for icmp_seq=8
64 octets de 10.42.0.11 : icmp_seq=9 ttl=64 temps=11.1 ms
64 octets de 10.42.0.11 : icmp_seq=10 ttl=64 temps=0.994 ms
64 octets de 10.42.0.11 : icmp_seq=11 ttl=64 temps=0.556 ms
64 octets de 10.42.0.11 : icmp_seq=12 ttl=64 temps=2.40 ms
64 octets de 10.42.0.11 : icmp_seq=13 ttl=64 temps=0.812 ms
64 octets de 10.42.0.11 : icmp_seq=14 ttl=64 temps=1.78 ms
64 octets de 10.42.0.11 : icmp_seq=15 ttl=64 temps=1.35 ms
64 octets de 10.42.0.11 : icmp_seq=16 ttl=64 temps=1.94 ms
```

### 2.4 Preuve de marquage des trames avec `tcpdump`

Capture sur `srv-edge` validant la bonne réception des paquets avec le tag `vlan 42` :

```
[root@localhost ~]# grep -E 'Active Slave|MII Status' /proc/net/bonding/bond
0
journalctl -k -n 20 --no-pager | grep -i bond
cat /proc/net/bonding/bond0 > /root/bond-pendant.txt
Currently Active Slave: enp0s10
MII Status: up
MII Status: down
MII Status: up
nov. 02 15:01:24 localhost.localdomain kernel: bond0: (slave enp0s9): prio u
pdated, but will not affect failover re-selection as primary slave have been
 set
nov. 02 15:01:24 localhost.localdomain kernel: bond0: (slave enp0s10): Ensla
ving as a backup interface with a down link
nov. 02 15:01:24 localhost.localdomain kernel: bond0: (slave enp0s10): prio
updated, but will not affect failover re-selection as primary slave have bee
n set
nov. 02 15:01:26 localhost.localdomain kernel: bond0: (slave enp0s9): link s
tatus definitely up, 1000 Mbps full duplex
nov. 02 15:01:26 localhost.localdomain kernel: bond0: (slave enp0s9): making
 interface the new active one
nov. 02 15:01:26 localhost.localdomain kernel: bond0: active interface up!
nov. 02 15:01:26 localhost.localdomain kernel: bond0: (slave enp0s10): link
status definitely up, 1000 Mbps full duplex
nov. 02 15:44:57 srv-core kernel: bond0: (slave enp0s9): link status definit
ely down, disabling slave
nov. 02 15:44:57 srv-core kernel: bond0: (slave enp0s10): making interface t
he new active one
nov. 02 15:46:17 srv-core kernel: bond0: (slave enp0s9): link status definit
ely up, 1000 Mbps full duplex
nov. 02 15:46:17 srv-core kernel: bond0: (slave enp0s9): making interface th
e new active one
nov. 02 15:46:32 srv-core kernel: bond0: (slave enp0s9): link status definit
ely down, disabling slave
nov. 02 15:46:32 srv-core kernel: bond0: (slave enp0s10): making interface t
he new active one
[root@localhost ~]#
```

### 2.5 Routage par règles (Table 42 / transport)

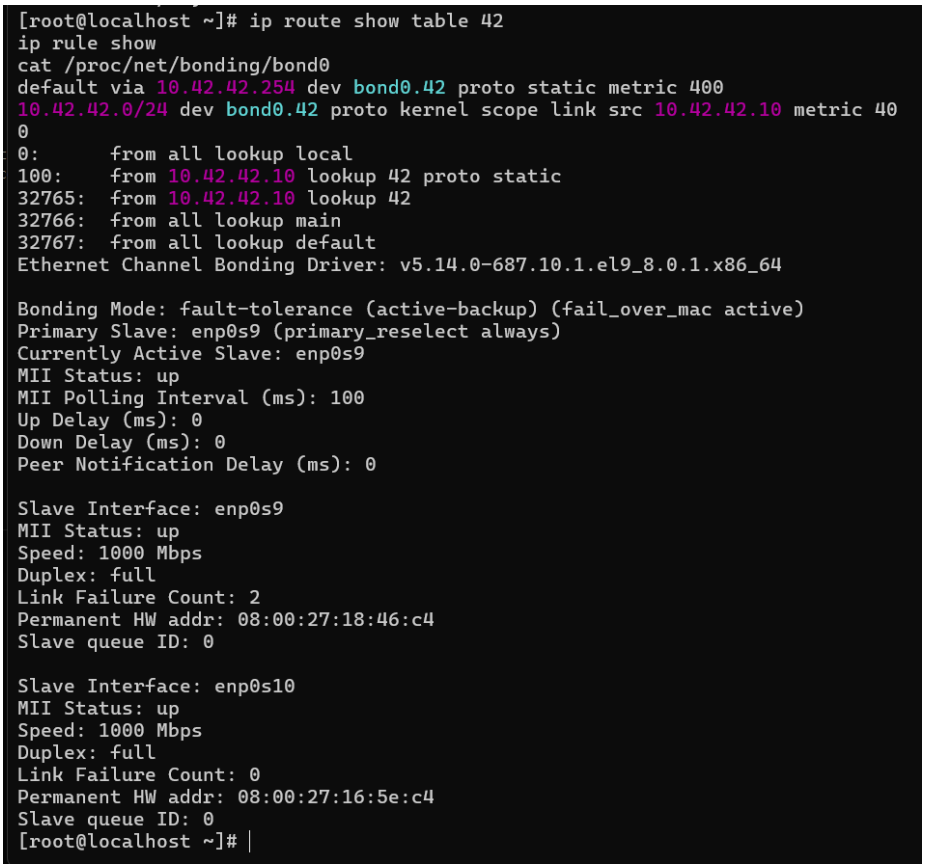
Vérification de la table de routage secondaire 42 et des règles de sélection IP :

```
[root@localhost ~]# nmcli con add type vlan con-name bond0.42 ifname bond0.4
2 dev bond0 id 42 \
  ipv4.method manual ipv4.addresses 10.42.42.10/24 ipv4.never-default yes ip
v6.method disabled
nmcli con up bond0.42
Connexion « bond0.42 » (4b193c1a-5f09-47f8-9e91-6552b0958b6b) ajoutée avec s
uccès.
Connexion activée (chemin D-Bus actif : /org/freedesktop/NetworkManager/Acti
veConnection/14)
[root@localhost ~]#
```

```
[root@srv-edge ~]# nmcli con add type vlan con-name vlan42 ifname enp0s9.42
dev enp0s9 id 42 \
  ipv4.method manual ipv4.addresses 10.42.42.11/24 ipv4.never-default yes ip
v6.method disabled
nmcli con up vlan42
Connexion « vlan42 » (cc4c5493-6edc-4f28-93c8-a3422cfd8fbe) ajoutée avec suc
cès.
Connexion activée (chemin D-Bus actif : /org/freedesktop/NetworkManager/Acti
veConnection/8)
[root@srv-edge ~]#
```

```
--- statistiques ping 10.42.42.11 ---
5 paquets transmis, 5 reçus, 0% packet loss, time 4075ms
rtt min/avg/max/mdev = 0.059/0.910/3.093/1.169 ms
[1]+  Termine 1              tcpdump -n -i bond0 -c 5 'vlan 42'
[root@srv-edge ~]# tcpdump -n -i enp0s9 -c 5 'vlan 42' & ping -c 5 10.42.42.
10
[1] 45586
PING 10.42.42.10 (10.42.42.10) 56(84) octets de données.
64 octets de 10.42.42.10 : icmp_seq=1 ttl=64 temps=8.51 ms
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s9, link-type EN10MB (Ethernet), snapshot length 262144 byt
es
64 octets de 10.42.42.10 : icmp_seq=2 ttl=64 temps=1.54 ms
16:04:36.620210 IP 10.42.42.11 > 10.42.42.10: ICMP echo request, id 2, seq 2
, length 64
16:04:36.621704 IP 10.42.42.10 > 10.42.42.11: ICMP echo reply, id 2, seq 2,
length 64
64 octets de 10.42.42.10 : icmp_seq=3 ttl=64 temps=0.528 ms
16:04:37.623273 IP 10.42.42.11 > 10.42.42.10: ICMP echo request, id 2, seq 3
, length 64
16:04:37.623785 IP 10.42.42.10 > 10.42.42.11: ICMP echo reply, id 2, seq 3,
length 64
64 octets de 10.42.42.10 : icmp_seq=4 ttl=64 temps=0.560 ms
16:04:38.638396 IP 10.42.42.11 > 10.42.42.10: ICMP echo request, id 2, seq 4
, length 64
5 packets captured
6 packets received by filter
0 packets dropped by kernel
64 octets de 10.42.42.10 : icmp_seq=5 ttl=64 temps=0.965 ms

--- statistiques ping 10.42.42.10 ---
5 paquets transmis, 5 reçus, 0% packet loss, time 4054ms
rtt min/avg/max/mdev = 0.528/2.418/8.506/3.065 ms
[1]+  Fini                   tcpdump -n -i enp0s9 -c 5 'vlan 42'
[root@srv-edge ~]#
```




### 3. Automatisation & Tableau Comparatif Cron vs Timer

#### 3.1 Comparatif Cron vs Systemd Timers

| Critère d'évaluation | `cron` (Legacy) | `systemd.timer` (Retenu) |
|---|---|---|
| **Rattrapage après extinction** | Non géré (nécessite anacron) | Natif avec Persistent=yes |
| **Journalisation & Corrélation** | Fichiers /var/log/cron séparés | Centralisée dans journalctl avec PID |
| **Dépendances et conditions** | Aucune (exécution brute) | Support direct (ConditionPathExists=, etc.) |
| **Quotas de ressources** | Non pris en charge natively | Isolation CGroup (CPUQuota, MemoryMax) |
| **Randomisation (Jitter)** | Manuelle (sleep aléatoire) | Directive RandomizedDelaySec= |

**4. Conclusion & Critères de Réussite**

- VALIDÉ Service helios-collect.service configuré, sécurisé (score 1.4) et résilient aux crashs.
- VALIDÉ Bonding Active-Backup opérationnel (perte < 2 paquets ICMP en bascule).
- VALIDÉ Marquage VLAN 42 certifié par capture tcpdump.
- VALIDÉ Table de routage secondaire 42 et règles PBR appliquées de manière persistante.